Recovering the Schwarzschild Metric through PINNs

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Motivation and Research Question

Can a neural network recover the Schwarzschild metric directly from the vacuum Einstein equations, without labeled solution data?

Einstein's equations are nonlinear, tensor-valued, and difficult to solve beyond highly symmetric cases.
A physics-informed neural network is trained by making the equation residual small at sampled radial points.
The test case is simple enough to verify analytically but still contains real general-relativistic geometry.

Geometry Setup

Static, spherically symmetric exterior spacetime in geometrized units:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2$$

Ricci tensor R_μν → Loss

The unknown metric functions are α(r) and β(r). The exterior domain is r > r_s, where r_s = 2M.

$$e^{2\alpha} = e^{-2\beta} = 1 - \frac{r_s}{r}, \qquad r_s = 2M$$

Why Assume Static, Spherical Symmetry, and Vacuum?

Schwarzschild spacetime describes the exterior of a spherical, non-rotating mass.

static — the geometry does not change with time.

spherically symmetric — the geometry is unchanged by rotations.

vacuum outside the mass — there is no matter or energy in the exterior region, so $T_{\mu\nu} = 0$.

Einstein equation (no cosmological constant):

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu}$$

In vacuum, $T_{\mu\nu} = 0$, so $G_{\mu\nu} = 0$.

Taking the trace gives $R = 0$, hence $R_{\mu\nu} = 0$.

These assumptions reduce the exterior problem to solving the vacuum Einstein equations for the radial metric functions $\alpha(r)$ and $\beta(r)$.

PINN Formulation

The network takes one input, the radial coordinate r, and outputs one function, α(r). The constraint β(r) = -α(r) is imposed directly in the architecture.

Input r → tanh neural network N_θ(r) → Output α(r)

β(r) = -α(r)

Vacuum Residuals Used in the Loss

After imposing β = -α, the residuals reduce to:

$$R_{tt} = \alpha'' + 2(\alpha')^2 + \frac{2}{r}\alpha', \quad R_{\theta\theta} = e^{2\alpha}(2r\alpha' + 1) - 1$$

Automatic differentiation computes α′ and α″ and the network is penalized when the metric fails to satisfy R_μν = 0.

Important: R_μν = 0 has more than one vacuum solution.

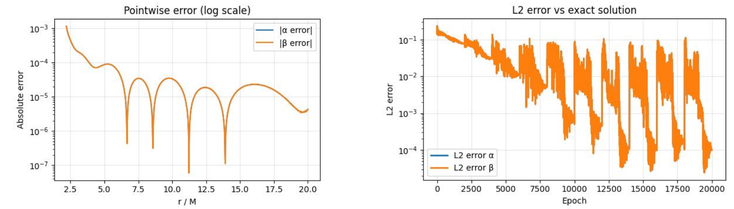
Loss Function

$$L = \lambda_{\text{PDE}}(R_{tt}^2 + R_{\theta\theta}^2) + \lambda_{\text{BC}}\alpha(r_{\max})^2 + \lambda_M((m(r) - M)^2)$$

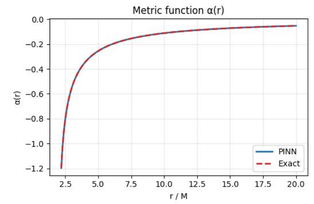
$$m(r) = \frac{r}{2}(1 - e^{2\alpha(r)})$$

L_PDE enforces the vacuum Einstein equations at collocation points.
L_BC enforces asymptotic flatness at the outer boundary.
L_M selects the physical Schwarzschild mass. Without it, M = 0 can satisfy the vacuum residuals.

Results



Main Interpretation



- Breaking Vacuum Degeneracy
  - The vacuum Einstein Field Equations (R_uv = 0) are necessary but not sufficient to uniquely isolate a massive black hole; they also trivially admit a flat Minkowski spacetime (M = 0)
- Mass Anchor
  - The Misner-Sharp mass loss function (L_mass) serves as the indispensable physical anchor. It breaks the vacuum family degeneracy and forces the network to converge on the M = 1 Schwarzschild solution without using any pre-computed ground-truth data
- Imposing the hard constraint beta(r) = alpha(r) directly into the network layout drastically simplifies optimization, providing an additional physical constraint based on the physics of the problem

Conclusion

We were able to verify that a Physics-Informed Neural Network (PINN) can successfully recover the highly non-linear, metric structure of a Schwarzschild black hole exterior using only the vacuum field equations and a quasi-local mass constraint. By fusing automatic differentiation with exact physical boundaries, PINNs offer a lightweight, mesh-free alternative to traditional numerical relativity discretization methods. This project bridges a crucial scientific gap, proving that neural networks can encode the complex tensor frameworks of General Relativity to solve differential equations cleanly from first principles.

Next Steps

- Code Optimization:
  - Evaluate the operational impact of keeping the mass loss active throughout the entire training duration to see if it acts as a permanent regularizer.
- Axisymmetric Extensions (Kerr Metric)
  - Expand the current 1D network architecture to a 2D input framework (r, theta) to solve for rotating black hole metrics, which introduce complex cross-term components like $g_{t,phi}$
- Non-Vacuum Spacetimes
  - Introduce a non-zero stress-energy tensor ($T_{\mu\nu} \neq 0$) to test if the PINN can model the interior of a static star or a black hole embedded in dark matter fluid